

Naive Bayesian classifier

Bayesian Classification:

Why?

- A statistical classifier: performs *probabilistic prediction, i.e.*, predicts class membership probabilities
- Foundation: Based on Bayes' Theorem.
- Performance: A simple Bayesian classifier, *naïve Bayesian classifier*, has comparable performance with decision tree and selected neural network classifiers
- Incremental: Each training example can incrementally increase/decrease the probability that a hypothesis is correct — prior knowledge can be combined with observed data

Bayes' Rule

$$p(h | d) = \frac{P(d | h)P(h)}{P(d)}$$

Understanding Bayes' rule

d = data

h = hypothesis (model)

- rearranging

$$p(h | d)P(d) = P(d | h)P(h)$$

$$P(d, h) = P(d, h)$$

the same joint probability

on both sides

Who is who in Bayes' rule

$P(h)$: prior belief (probability of hypothesis h before seeing any data)

$P(d | h)$: likelihood (probability of the data if the hypothesis h is true)

$P(d) = \sum_h P(d | h)P(h)$: data evidence (marginal probability of the data)

$P(h | d)$: posterior (probability of hypothesis h after having seen the data d)

Example of Bayes Theorem

- Given:
 - A doctor knows that meningitis causes stiff neck 50% of the time
 - Prior probability of any patient having meningitis is 1/50,000
 - Prior probability of any patient having stiff neck is 1/20
- If a patient has stiff neck, what's the probability he/she has meningitis?

$$P(M | S) = \frac{P(S | M)P(M)}{P(S)} = \frac{0.5 \times 1/50000}{1/20} = 0.0002$$

Choosing Hypotheses

- *Maximum Likelihood* hypothesis:

$$h_{ML} = \arg \max_{h \in H} P(d | h)$$

$$h_{MAP} = \arg \max_{h \in H} P(h | d)$$

- Generally we want the most probable hypothesis given training data. This is the *maximum a posteriori* hypothesis:
 - Useful observation: it does not depend on the denominator $P(d)$

Bayesian Classifiers

- Consider each attribute and class label as random variables
- Given a record with attributes $(A_1, A_2, \dots, A_n)$
 - Goal is to predict class C
 - Specifically, we want to find the value of C that maximizes $P(C | A_1, A_2, \dots, A_n)$
- Can we estimate $P(C | A_1, A_2, \dots, A_n)$ directly from data?

Bayesian Classifiers

- Approach:
 - compute the posterior probability $P(C \mid A_1, A_2, \dots, A_n)$ for all values of C using the Bayes theorem

$$P(C \mid A_1 A_2 \dots A_n) = \frac{P(A_1 A_2 \dots A_n \mid C) P(C)}{P(A_1 A_2 \dots A_n)}$$

- Choose value of C that maximizes $P(C \mid A_1, A_2, \dots, A_n)$
 - Equivalent to choosing value of C that maximizes $P(A_1, A_2, \dots, A_n \mid C) P(C)$
- How to estimate $P(A_1, A_2, \dots, A_n \mid C)$?

Naïve Bayes Classifier

- Assume independence among attributes A_i when class is given:
 - $P(A_1, A_2, \dots, A_n | C) = P(A_1 | C) P(A_2 | C) \dots P(A_n | C)$
 - Can estimate $P(A_i | C)$ for all A_i and C .
 - This is a simplifying assumption which may be violated in reality
- The Bayesian classifier that uses the Naïve Bayes assumption and computes the MAP hypothesis is called Naïve Bayes classifier

$$c_{Naive Bayes} = \arg \max_c P(c)P(\mathbf{x} | c) = \arg \max_c P(c) \prod_i P(a_i | c)$$

How to Estimate Probabilities from Data?

$$P(C) = N_C / N$$

– e.g., $P(\text{No}) = 7/10$,
 $P(\text{Yes}) = 3/10$

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- For discrete attributes:

$$P(A_i | C_k) = |A_{ik}| / N_C$$

- where $|A_{ik}|$ is number of instances having attribute A_i and belongs to class C_k
- Examples:

$$P(\text{Status}=\text{Married}|\text{No}) = 4/7$$

$$P(\text{Refund}=\text{Yes}|\text{Yes})=0$$

How to Estimate Probabilities from Data?

- For continuous attributes:
 - **Discretize** the range into bins
 - one ordinal attribute per bin
 - violates independence assumption
 - **Two-way split:** $(A < v)$ or $(A > v)$
 - choose only one of the two splits as new attribute
 - **Probability density estimation:**
 - Assume attribute follows a normal distribution
 - Use data to estimate parameters of distribution (e.g., mean and standard deviation)
 - Once probability distribution is known, can use it to estimate the conditional probability $P(A_i|c)$

How to Estimate Probabilities

from Data?

Tid	Refund	Marital Status	Taxable Income	Evade
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Normal distribution:

$$P(A_i | c_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} e^{-\frac{(A_i - \mu_{ij})^2}{2\sigma_{ij}^2}}$$

– One for each (A_i, c_i) pair

- For (Income, Class=No):

– If Class=No

- sample mean = 110
- sample variance = 2975

$$P(\text{Income} = 120 | \text{No}) = \frac{1}{\sqrt{2\pi(54.54)}} e^{-\frac{(120-110)^2}{2(2975)}} = 0.0072$$

Naïve Bayesian Classifier:

Training Dataset

Class:

C1:buys_computer = 'yes'

C2:buys_computer = 'no'

New Data:

X = (age <=30,

Income = medium,

Student = yes

Credit_rating = Fair)

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Naïve Bayesian Classifier: An Example

Given X (age=youth, income=medium, student=yes, credit=fair)

Maximize $P(X|C_i)P(C_i)$, for $i=1,2$

First step: Compute $P(C)$ The prior probability of each class can be computed based on the training tuples:

$$P(\text{buys_computer}=\text{yes})=9/14=0.643$$

$$P(\text{buys_computer}=\text{no})=5/14=0.357$$

Naïve Bayesian Classifier:

An Example

Given X (age=youth, income=medium, student=yes, credit=fair)

Maximize $P(X|C_i)P(C_i)$, for $i=1,2$

Second step: compute $P(X|C_i)$

$$\begin{aligned} P(X|\text{buys_computer=yes}) &= P(\text{age=youth}|\text{buys_computer=yes}) \times \\ &\quad P(\text{income=medium}|\text{buys_computer=yes}) \times \\ &\quad P(\text{student=yes}|\text{buys_computer=yes}) \times \\ &\quad P(\text{credit_rating=fair}|\text{buys_computer=yes}) \\ &= 0.044 \end{aligned}$$

$$P(\text{age=youth}|\text{buys_computer=yes}) = 0.222$$

$$P(\text{income=medium}|\text{buys_computer=yes}) = 0.444$$

$$P(\text{student=yes}|\text{buys_computer=yes}) = 6/9 = 0.667$$

$$P(\text{credit_rating=fair}|\text{buys_computer=yes}) = 6/9 = 0.667$$

Naïve Bayesian Classifier: An Example

Given X (age=youth, income=medium, student=yes, credit=fair)

Maximize $P(X|C_i)P(C_i)$, for $i=1,2$

Second step: compute $P(X|C_i)$

$$\begin{aligned} P(X|\text{buys_computer=no}) &= P(\text{age=youth}|\text{buys_computer=no}) \times \\ &\quad P(\text{income=medium}|\text{buys_computer=no}) \times \\ &\quad P(\text{student=yes}|\text{buys_computer=no}) \times \\ &\quad P(\text{credit_rating=fair}|\text{buys_computer=no}) \\ &= 0.019 \end{aligned}$$

$$P(\text{age=youth}|\text{buys_computer=no}) = 3/5 = 0.666$$

$$P(\text{income=medium}|\text{buys_computer=no}) = 2/5 = 0.400$$

$$P(\text{student=yes}|\text{buys_computer=no}) = 1/5 = 0.200$$

$$P(\text{credit_rating=fair}|\text{buys_computer=no}) = 2/5 = 0.400$$

Naïve Bayesian Classifier: An Example

Given X (age=youth, income=medium, student=yes, credit=fair)

Maximize $P(X|C_i)P(C_i)$, for $i=1,2$

We have computed in the first and second steps:

$$P(\text{buys_computer}=\text{yes})=9/14=0.643$$

$$P(\text{buys_computer}=\text{no})=5/14=0.357$$

$$P(X|\text{buys_computer}=\text{yes})= 0.044$$

$$P(X|\text{buys_computer}=\text{no})= 0.019$$

Third step: compute $P(X|C_i)P(C_i)$ for each class

$$P(X|\text{buys_computer}=\text{yes})P(\text{buys_computer}=\text{yes})=0.044 \times 0.643=0.028$$

$$P(X|\text{buys_computer}=\text{no})P(\text{buys_computer}=\text{no})=0.019 \times 0.357=0.007$$

The naïve Bayesian Classifier predicts **X belongs to class**
("buys_computer = yes")

Example

Training set :
(Öğrenme Kümesi)

<i>Tid</i>	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
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3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
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Given a Test Record:

$X = (\text{Refund} = \text{No}, \text{Married}, \text{Income} = 120\text{K})$

Example of Naïve Bayes Classifier

Given a Test Record:

$$X = (\text{Refund} = \text{No}, \text{Married}, \text{Income} = 120\text{K})$$

naive Bayes Classifier:

$P(\text{Refund}=\text{Yes}|\text{No}) = 3/7$
 $P(\text{Refund}=\text{No}|\text{No}) = 4/7$
 $P(\text{Refund}=\text{Yes}|\text{Yes}) = 0$
 $P(\text{Refund}=\text{No}|\text{Yes}) = 1$
 $P(\text{Marital Status}=\text{Single}|\text{No}) = 2/7$
 $P(\text{Marital Status}=\text{Divorced}|\text{No})=1/7$
 $P(\text{Marital Status}=\text{Married}|\text{No}) = 4/7$
 $P(\text{Marital Status}=\text{Single}|\text{Yes}) = 2/7$
 $P(\text{Marital Status}=\text{Divorced}|\text{Yes})=1/7$
 $P(\text{Marital Status}=\text{Married}|\text{Yes}) = 0$

For taxable income:

If class=No: sample mean=110
 sample variance=2975
If class=Yes: sample mean=90
 sample variance=25

- $P(X|\text{Class}=\text{No}) = P(\text{Refund}=\text{No}|\text{Class}=\text{No})$
 $\times P(\text{Married}|\text{Class}=\text{No})$
 $\times P(\text{Income}=120\text{K}|\text{Class}=\text{No})$
 $= 4/7 \times 4/7 \times 0.0072 = 0.0024$
- $P(X|\text{Class}=\text{Yes}) = P(\text{Refund}=\text{No}|\text{Class}=\text{Yes})$
 $\times P(\text{Married}|\text{Class}=\text{Yes})$
 $\times P(\text{Income}=120\text{K}|\text{Class}=\text{Yes})$
 $= 1 \times 0 \times 1.2 \times 10^{-9} = 0$

Since $P(X|\text{No})P(\text{No}) > P(X|\text{Yes})P(\text{Yes})$

Therefore $P(\text{No}|X) > P(\text{Yes}|X)$
 $\Rightarrow \text{Class} = \text{No}$

Naïve Bayes (Summary)

- Advantage
 - Robust to isolated noise points
 - Handle missing values by ignoring the instance during probability estimate calculations
 - Robust to irrelevant attributes
- Disadvantage
 - Assumption: class conditional independence, which may cause loss of accuracy
 - Independence assumption may not hold for some attribute. Practically, dependencies exist among variables
 - Use other techniques such as Bayesian Belief Networks (BBN)

Remember

- Bayes' rule can be turned into a classifier
- Maximum A Posteriori (MAP) hypothesis estimation incorporates prior knowledge; Max Likelihood (ML) doesn't
- Naive Bayes Classifier is a simple but effective Bayesian classifier for vector data (i.e. data with several attributes) that assumes that attributes are independent given the class.
- Bayesian classification is a generative approach to classification